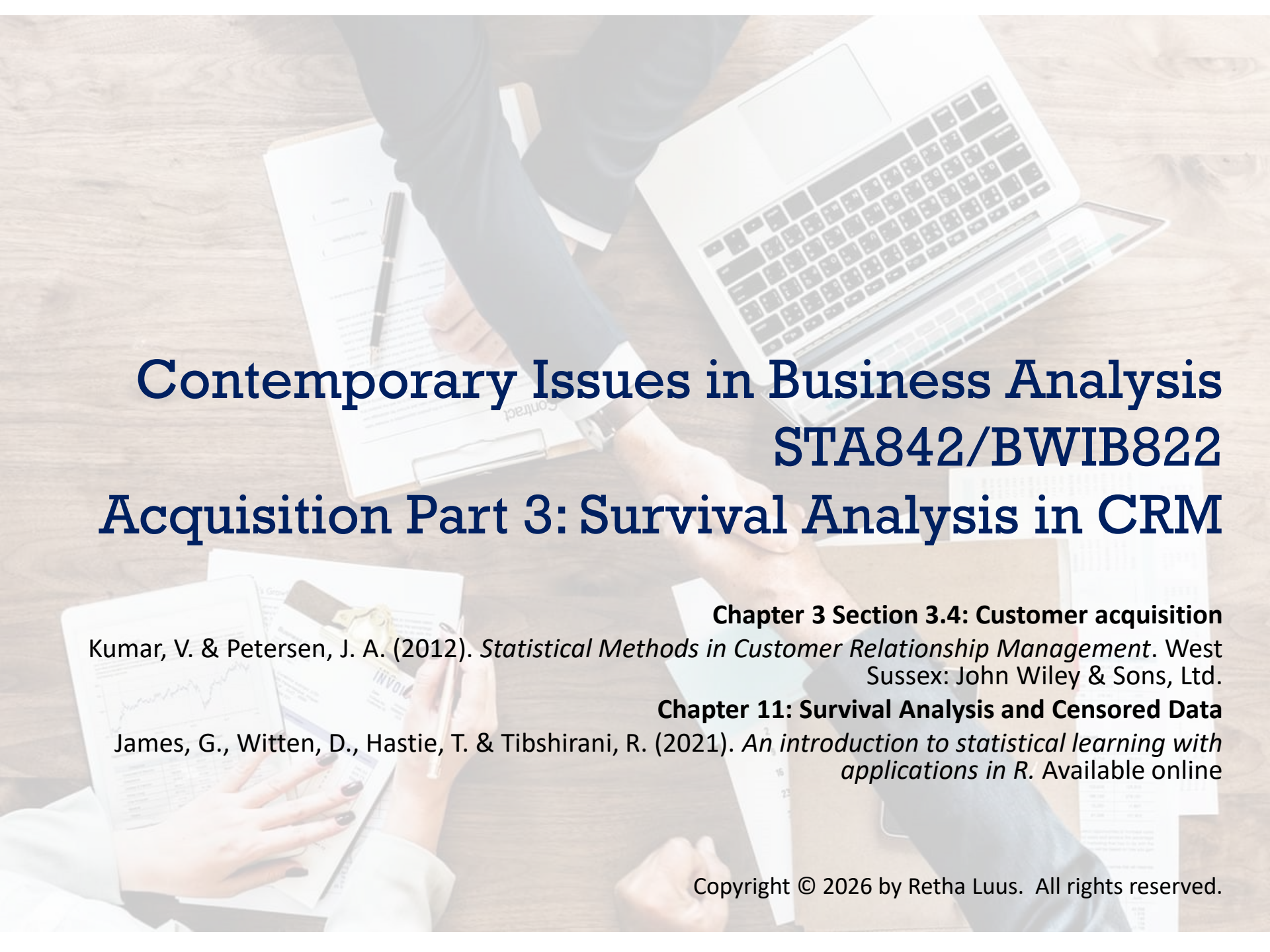




Contemporary Issues in Business Analysis

STA842/BWIB822

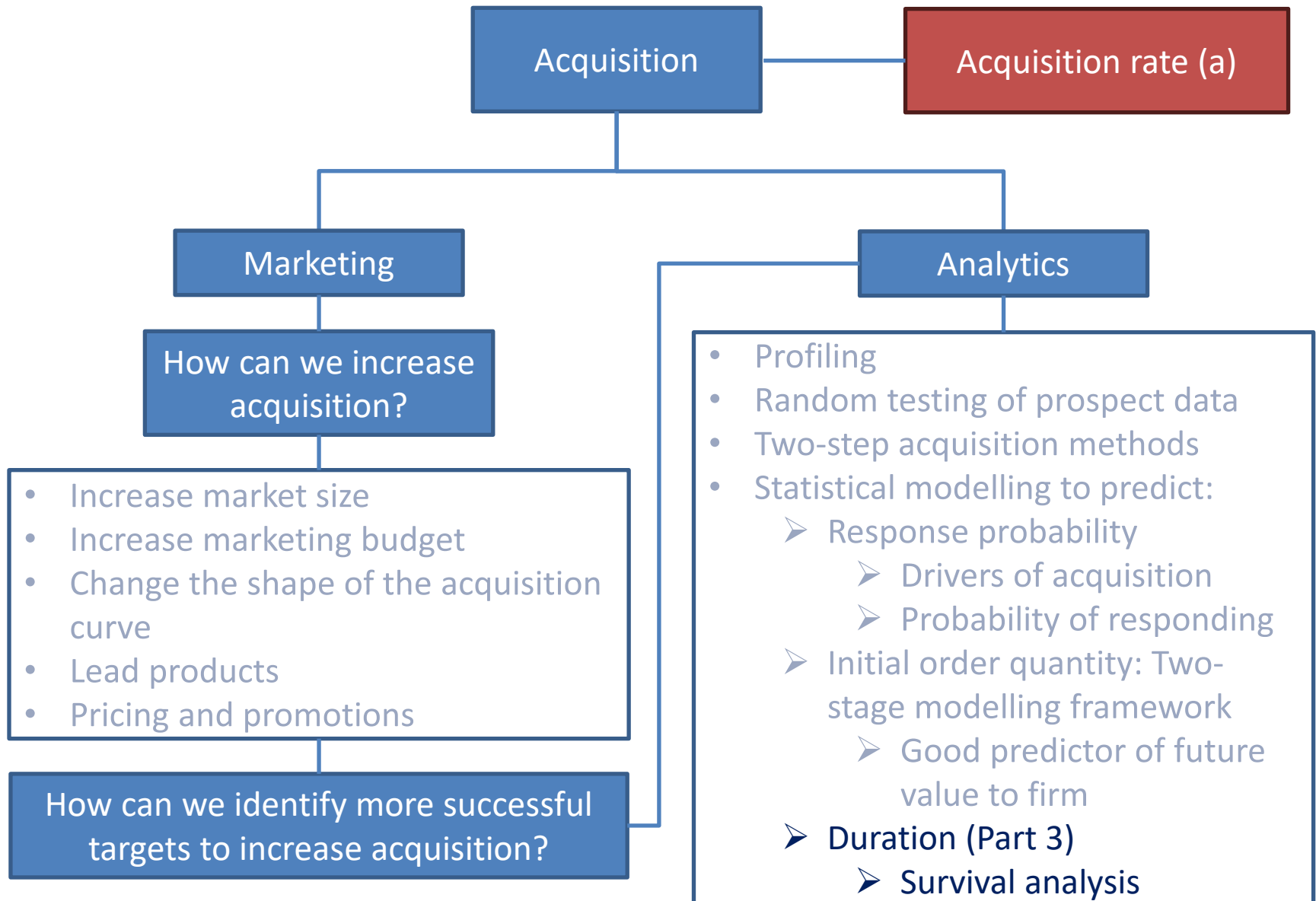
Acquisition Part 3: Survival Analysis in CRM

Chapter 3 Section 3.4: Customer acquisition

Kumar, V. & Petersen, J. A. (2012). *Statistical Methods in Customer Relationship Management*. West Sussex: John Wiley & Sons, Ltd.

Chapter 11: Survival Analysis and Censored Data

James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021). *An introduction to statistical learning with applications in R*. Available online

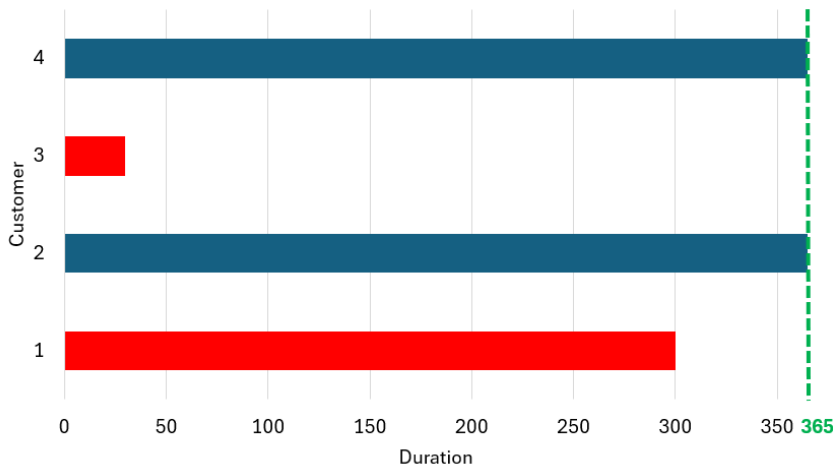


Introduction

- **Acquisition** about **getting good customers** for the business
- **Retention** (next): interventions to hold onto good customers as long as possible
 - **Core question**: We spent \$50 to acquire them, **will we make money?**
 - **To answer** this, we require **Margin * Duration**
 - Without duration (or tenure), return on investment (ROI) is unknown
- **Analyse duration** of customer relationship with firm
 - **Or**: predict **tenure** (time until the customer will leave their business)
- **Response** to model: duration $\Rightarrow$ time $\Rightarrow$ numeric/continuous
- Model ubiquitous with numeric response: linear regression!
 - **For this application**, is linear regression the “**best**” model to use?

Introduction

Scenario: predict **duration**
(or tenure) using linear
regression



- **Blue**: customer is still active
- **Red**: customer has churned

- **In the first year**:
 - Some customers will leave the company (**churn - red**)
 - Others will still be with the company at day 365 (**active - blue**)
 - To predict duration using **linear regression** you'll need the **observed/measured duration** of the customer/firm relationship
 - Only model on customers that have churned
 - Ignoring customers that haven't churned might **discard valuable information** (selection bias)
- Linear regression not the “best” model to use

Introduction

- **Scenario**: predict **churn** using classification model

Customer	Status	Duration (days)
1	Churned	300
2	Active	365
3	Churned	30
4	Active	365

- Suppose it takes 6 months (± 180 days) to recover new customer's acquisition expense
 - Customer 3 churned after 30 days
 - Customer 1 churned after 300 days
- A classification model treats them as identical risks, even though the company at least recovered Customer 1's cost
- Excluding duration is another form of information loss

Introduction

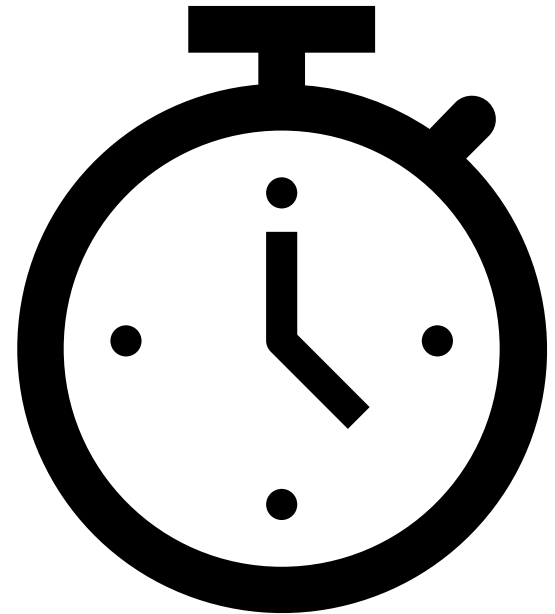
- **Problem**: we need to analyse **time-to-event** data
- **Survival analysis** is a collection of tools for analysing this data
- **Today's roadmap**:
 - **Understanding** time-to-event data
 - **Describing** time-to-event data
 - **Modelling** time-to-event data

TIME-TO-EVENT DATA

Definition

Censoring

Notation



Time-to-Event Data

Time-to-event data is data on how long it takes until some event occurs

Example: insurance policy lapse

- Predict how long after inception the client lapses on their policy
- Event of interest: lapse
- Time component: duration of client-firm relationship (response variable)

Example: study customer repurchase behaviour in retail

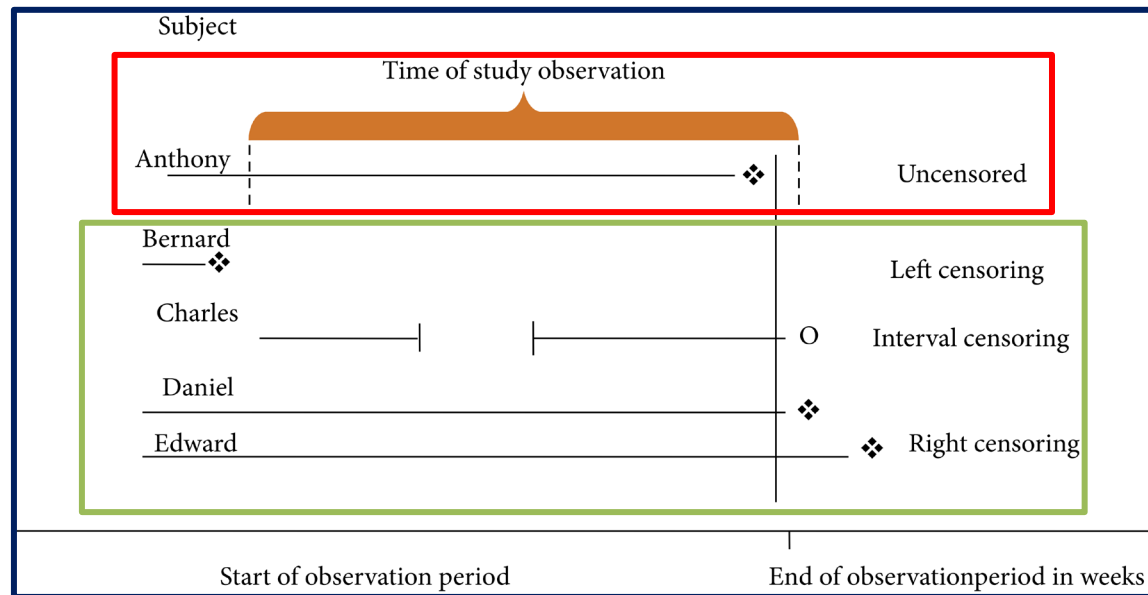
- Predict how long until a customer makes their next purchase
- Event of interest: purchase
- Time component: duration until next purchase (response variable)

Time-to-Event Data

- Yes, *time* or *duration* is the response of interest
- **Focus** of survival analysis: examining the **probability of an event occurring** at time t
- If **all** observation units (customers) **had the event** $\Rightarrow$ use **standard statistical methods**
- Survival data collected in an **observation window**
 - Some observation units will experience the event (**time-to-event known**)
 - Others will not have experienced the event by the end of the study (**time-to-event unknown**)
- This touches on the concept of **censoring**

Censoring

- **Censoring** occurs when we have some information about individual survival time, but we do not know the survival time exactly
- **Anthony's** event is measured **before the end of the observation window** is reached
⇒ **uncensored** observation



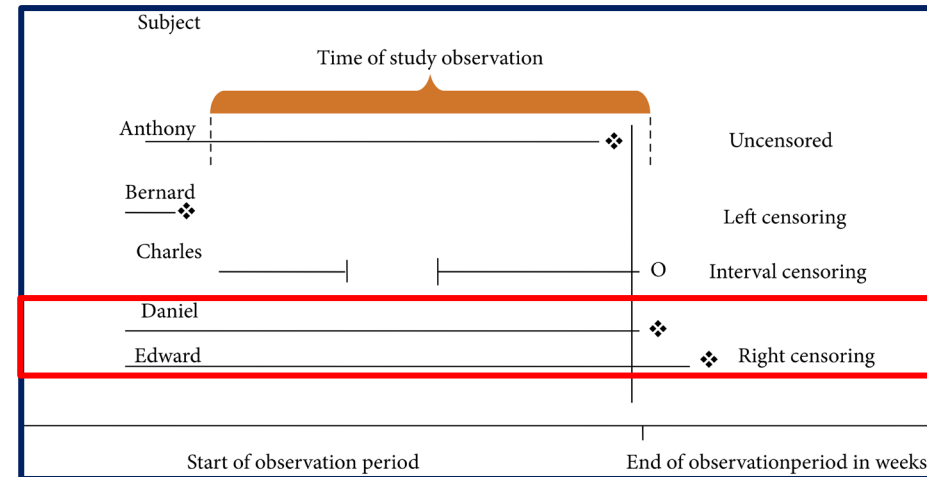
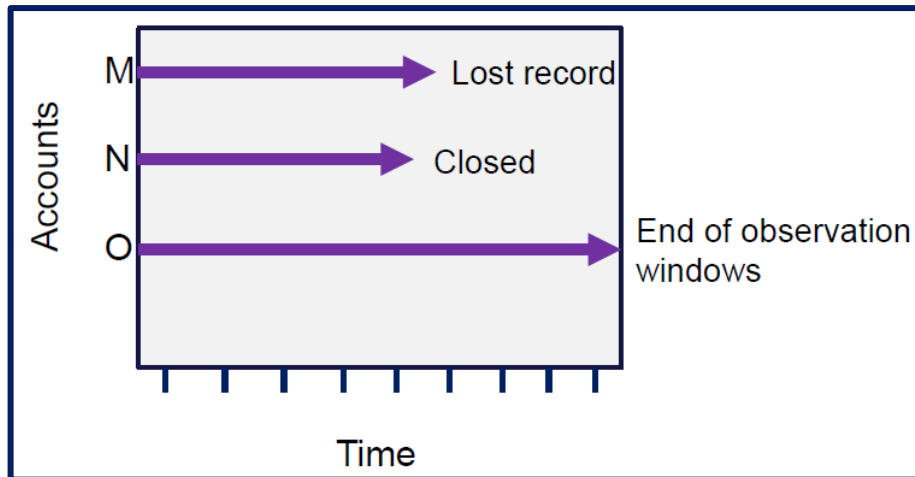
(Turkson, et al., 2021)

- Records that **haven't experience the event** by the end of your observation window are **censored**

Three types of censoring:
Right censoring, **Left** censoring, **Interval** censoring

Right Censoring

- An observation is right-censored if the **end of the observation window is reached** before the event occurs

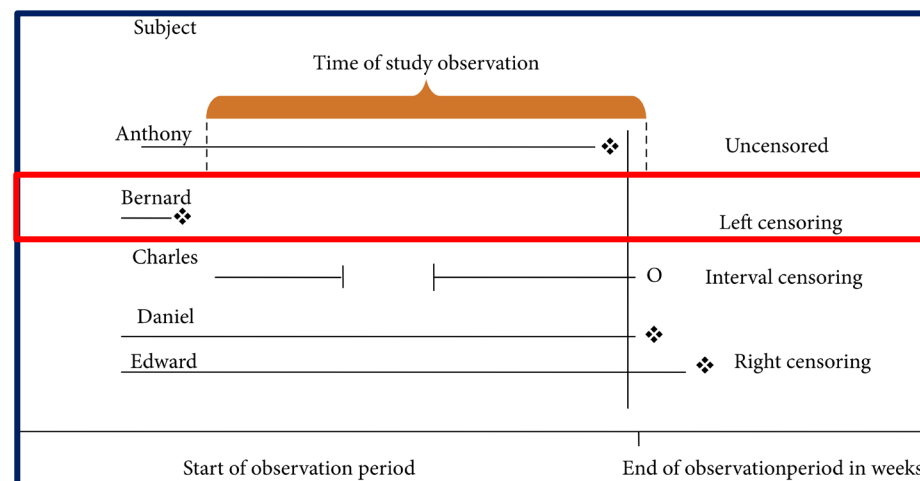
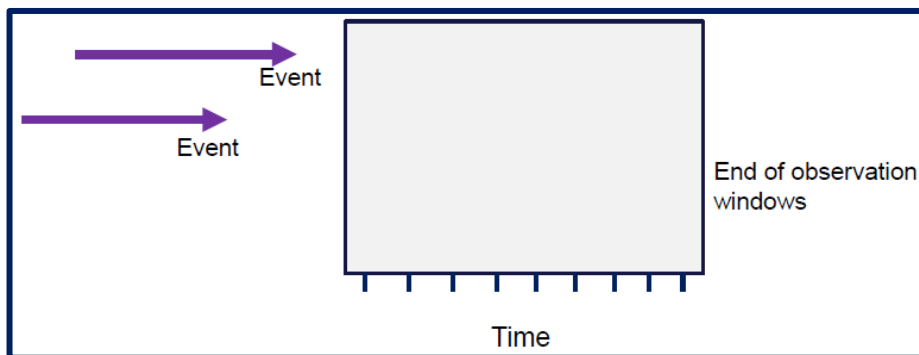


(Turkson, et al., 2021)

- Possible **reasons**:
 - The record “lives” beyond the end of the observation window
 - Record can’t be located
 - Drop out before the investigation/analysis is completed
- This describes approx. 90% of CRM data (active customers) – your focus

Left Censoring

- An observation is left-censored if the observation experiences the event **before the start** of the observation window

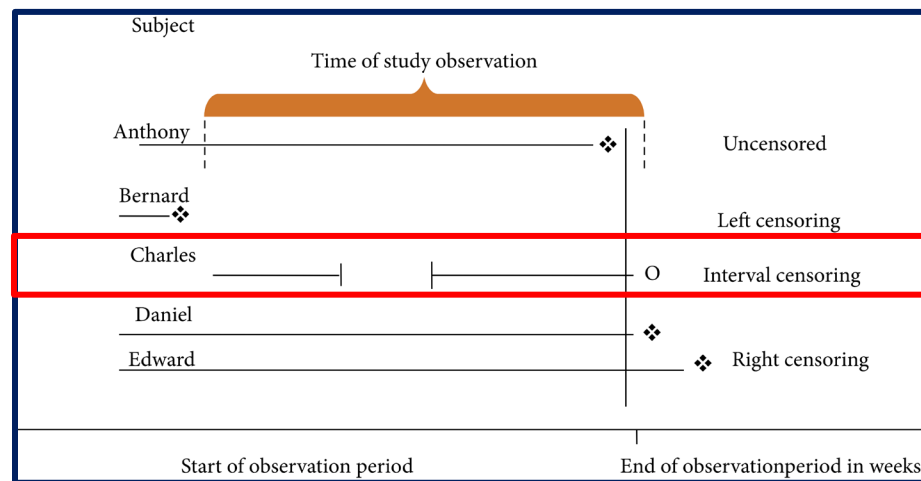
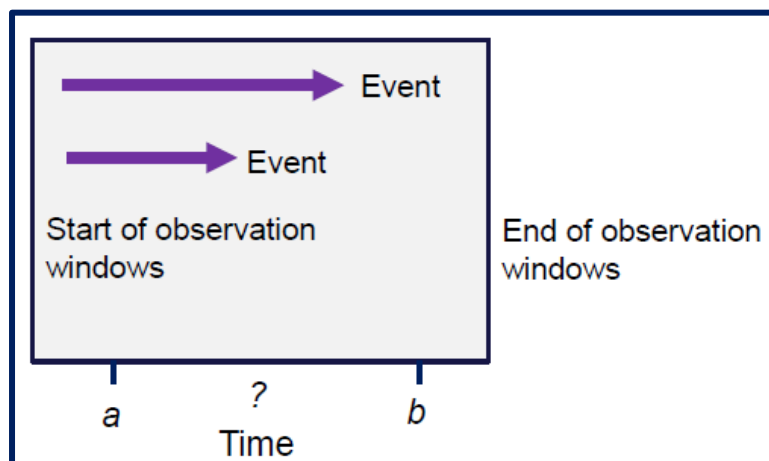


(Turkson, et al., 2021)

- Possible **reasons**:
 - In a medical study someone dies before the drug trial begins
- Rare in CRM**

Interval Censoring

- If the **information** concerning the event's timing is only **partially known** $\Rightarrow$ event took place between time a and b



(Turkson, et al., 2021)

- Possible **reasons**:
 - See a patient once a week to determine whether the event has occurred or not and somewhere between visits the event occurred
 - Common in subscriptions checked monthly

Censoring Last Remarks

Ignoring censoring is not advisable:

- **biased** results
 - Not considering records that “survived” past the observation window skews perception by only looking at records that have experienced the event
- **less efficient** estimation
 - Started with, e.g., 5000 records while only 2000 records experienced event
 - Ignoring censoring throws away 60% of valuable informative information

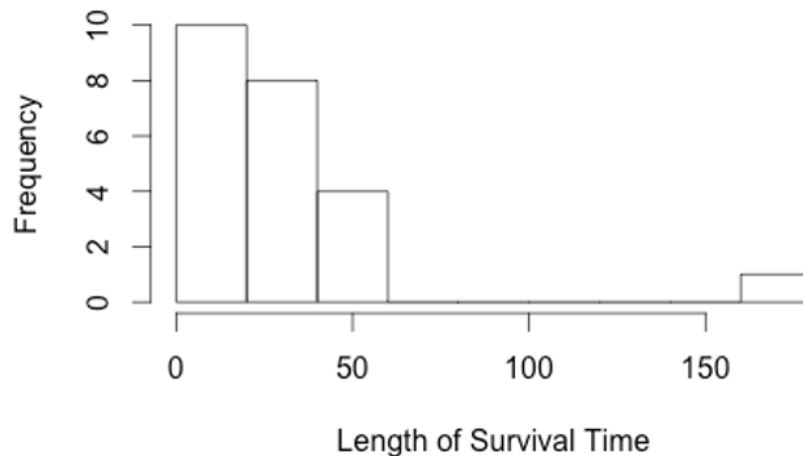
Important assumption in survival analysis: censoring unrelated to survival time

- **Uninformative censoring**
- **Examples:**
 - observations lost to follow-up should be as likely to experience the event of interest as those observations still part of the study
 - when the reasons for termination are unrelated to the risk of the event or out of the control of the investigator

Time-to-Event Distributional Shape

- Survival data are **rarely normally distributed**
- Instead, survival data tends to be **positively skewed**

Histogram of Survival Time in AML Patients



- Many events observed early in the study
- Fewer events observed later

(https://sphweb.bumc.bu.edu/otlt/MPH-Modules/BS/R/R7_LogisticRegression-Survival/R7_LogisticRegression-Survival4.html)

- For these reasons (**censoring**, **skewness**) it is necessary to use the **methods of survival analysis**

Time-to-Event Data: Continuous vs Discrete

- **Continuous-time survival analysis** is used when the measurement of time can be divided so that no two subjects can experience an event at the same time (i.e. no ties)
- **Discrete-time survival analysis** is used when the observational units of time are aggregate intervals (i.e. the precise time of the event is unknown)
- **Kumar and Petersen** apply **continuous-time survival analysis**

Survival Analysis Terminology

- Let T denote **survival time** (failure time or event time): $T \geq 0$
 - Time at which the event occurs (lapse, next purchase, default)
 - $T = 0$: origin (policy inception, acquisition, account opened)
- **Examples** of survival time:
 - Nr of months after which the client lapses on their policy
 - Nr of days until next purchase
 - Nr of months until loan default
- Let C denote **censoring time** (time at which censoring occurs)
- **Examples** of censoring time:
 - End of observation window during which payment behaviour monitored
 - End of retail observation window during which purchasing behaviour monitored
 - End of performance window during which client's good/bad behaviour monitored

Survival Analysis Terminology

- Only T or C will be observed for each observation in the study
 - **Example**: predict time until customers lapse on their policy
 - Either they **lapse within observation window**: measure time of lapse (T)
 - **Policy still active** at **end** of observation window (**censored**): measure censoring time (C)
- The following **random variable** is observed as the response for survival analysis:

$$Y = \min(T, C)$$

Survival Analysis Terminology

- Furthermore, you also want to note whether $Y = \min(T, C)$ is an **event time** or **censoring time**
- You could call this the **event** or **status indicator**:
 - It will take a **value of 1** when $Y = T$ (event time; $T \leq C$)
 - It will take a **value of 0** when $Y = C$ (censoring time; $T > C$)
- Let's denote it by δ and define it formally as:

$$\delta = \begin{cases} 1, & \text{event} \\ 0, & \text{censored} \end{cases}$$

⇒ **survival data** summarised into:
 $(Y, \delta, \underline{X})$

X: **covariates** measured in the study

Note definition

➤ This **required for R**
In SAS, you specify the
censoring level

Demo Data

- Simulate data for $N = 100$ customers
- **Feature engineering:**
 - **Segment**: 40% Premium, 60% Basic
 - **Age** $\sim N(\mu = 35, \sigma = 10)$
 - **Average spend** $\sim N(\mu = 500, \sigma = 100)$
- **Survival profile:**
 - **Hazard rates** (segment churn risk): Premium = 0.02, Basic = 0.05
 - **Event time** $T = \exp(1) /_{Hazard} (\text{True_Time})$
 - **Status** (δ): churned (=1) or active/censored (=0)
- **Constraints:**
 - **Observation window**: maximum observation period of 24 months
 - **Right censoring:**
 - If $T \geq 24$, customer marked as active/censored (Status=0, Tenure=24)
 - If $T < 24$, customer marked as churned (Status=1) at that specific time

Demo Data: Code

```
data customer_churn;
  call streaminit(12345);

do ID = 1 to 100; (Changed N=1000 for modelling results)
  if rand('Uniform') < 0.4 then Segment='Premium'; else Segment='Basic'; /* Segment */
  Avg_Spend = round(rand('Normal', 500, 100)); /* Avg_Spend */

  do until (Age >= 21); /* Age */
    Age = round(rand('Normal', 35, 10));
  end;

  if Segment='Premium' then Hazard=0.02; else Hazard=0.05; /* Hazard rate */
  True_Time = rand('Exponential') / Hazard; /* Time to event */

  if True_Time > 24 then do; Tenure=24; Status=0; end; /* Censored */
  else do; Tenure=round(True_Time, 0.1); Status=1; end; /* Churned */

output;
end;
run;
```

Demo Data: Snapshot

ID	Segment	Avg_Spend	Age	Hazard	True_Time	Tenure	Status
1	Basic	528	38	0.05	25.41886986	24	0
2	Basic	286	26	0.05	2.652700495	2.7	1
3	Basic	576	30	0.05	1.375330108	1.4	1
4	Premium	474	27	0.02	14.97474257	15	1
5	Premium	566	40	0.02	0.314930763	0.3	1

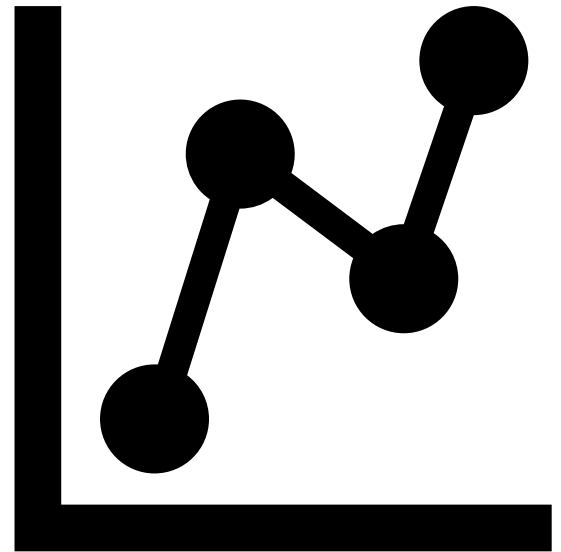
Variable	N	Mean	Std Dev	Minimum	Maximum
ID	100	50.5000000	29.0114920	1.0000000	100.0000000
Avg_Spend	100	490.3300000	99.8261170	197.0000000	709.0000000
Age	100	37.3100000	7.8953957	21.0000000	55.0000000
Hazard	100	0.0404000	0.0140648	0.0200000	0.0500000
True_Time	100	25.8731948	26.5443298	0.2215536	138.2801875
Tenure	100	15.5350000	8.6606200	0.2000000	24.0000000
Status	100	0.6400000	0.4824182	0	1.0000000

DESCRIBING TIME-TO-EVENT DATA

Survival function

Kaplan-Meier survival curve

Log-rank test



Describing Time-to-Event Data



Methods used to **describe** time-to-event data:

Kaplan-Meier survival curve
Stratified Kaplan-Meier survival curve
Log-rank test



These are considered **univariate analyses**

Look at the survival **on its own** or with respect to a **single factor** (stratified)

Familiar Characteristics

- Assume **continuous time**
- Probability density function (**pdf**)

$$f(t)$$

- **Density** around event of interest

$$\lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t)}{\Delta t}$$

- Cumulative distribution function (**cdf**)

$$F(t) = P(T \leq t)$$

- Unique to every random variable
- Pdf/cdf **relationship**: $f(t) = F'(t)$

Survival Function

- **Survival function** or **survival probability**: probability that a customer survives longer than time t
- Time origin **examples**:
 - Policy inception
 - Acquisition
 - Account opened
- Survival probabilities at different times t provide important information
 - **Describe survival experience** of study cohort

Survival Function

- In survival analysis you **won't be asking:**

probability that event occurs **before** some time (cdf $F(t)$)

- Here you are **interested in:**

probability of surviving **past** some time

- **Examples:**

- Probability that client lapses after **18 months**?
- Probability that next purchase is after **30 days**?
- Probability that client defaults after **6 months**?

Survival Function

- This defines the survival function:

$$S(t) = P(T > t) = 1 - F(t)$$

- Properties:
 - Starts at 1 ($S(0) = 1$)
 - Monotonically decreases to 0 ($S(t) \rightarrow 0$ as $t \rightarrow \infty$)
- Larger values of $S(t)$ imply that it is less likely the event occurs before time t
- Example: company wants to model relationship duration
 - T represents time customer leaves company
 - $S(t) = P(T > t)$ represents probability customer leaves after time t
 - Let $S(t) = 0.8$ where $t = 12$ (months) ($S(t) = P(T > 12) = 0.8$)
 - Small probability (0.2) of customer leaving within one year

Kaplan-Meier Survival Curve

- **Censoring complicates** estimation of survival function $S(t)$
- Well-known estimator of the survival function: **Kaplan-Meier estimator**

- Handles censoring by **adjusting the risk set** at every step

$$\hat{S}(d_k) = \prod_{j=1}^k \frac{r_j - q_j}{r_j}$$

- r_j : nr of observations alive just before time d_j (**at risk observations**)
- q_j : nr of observations that **experienced event** at time d_j
- $r_j - q_j$: nr of observations **survived beyond** time d_j
 - $\frac{r_j - q_j}{r_j}$: **proportion** of observations alive before time d_j that have **survived beyond** time d_j
- Plotting $\hat{S}(d_k)$ against time: **Kaplan-Meier (KM) survival curve**

Demo: Kaplan-Meier Survival Curve

```
proc lifetest data=customer_churn nelson  
    conftype=asinsqrt  
    plots=(survival (atrisk=0 to 24 by 5 test));  
    time tenure*status(0); /* Specify censoring  
level */  
run;
```

- **Note:**

- asinsqrt:
 - Forces survival probability confidence boundaries to stay between 0 and 1
 - Generally, more accurate than default method
- time: specifies survival response consisting of time and status
- status (0): specify the censoring level in brackets (0 in this example)

Demo: Kaplan-Meier Survival Curve

- **Summary statistics:**

Summary Statistics for Time Variable Tenure				
Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	.	ASINSQRT	.	.
50	18.0500	ASINSQRT	13.6000	22.3000
25	6.0000	ASINSQRT	4.6000	10.6000

- **Median survival time:** 18.05 months

- Approx: 18 months
- **Interpretation:** 50% of the customers are expected to end their relationship by 18 months, while the remaining 50% are expected to stay customers longer than 18 months

Demo: Kaplan-Meier Survival Curve

- Estimated survival probabilities at unique event time points:

The LIFETEST Procedure

Kaplan-Meier estimate

Nr at risk after time point t

T=0:

$$(r_0 - q_0)/r_0$$
$$= (100 - 0)/100 = 1$$

T=0.2:

$$(r_0 - q_0)/r_0$$
$$\times (r_{0.2} - q_{0.2})/r_{0.2}$$
$$= (1) \cdot (100 - 1)/100$$
$$= 0.99$$

Tenure	Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000	1.0000	0	0	0	100
0.2000	0.9900	0.0100	0.00995	1	99
0.3000	0.9800	0.0200	0.0140	2	98
0.5000	0.9700	0.0300	0.0171	3	97
1.1000	0.9600	0.0400	0.0196	4	96
1.4000	.	.	.	5	95
1.4000	0.9400	0.0600	0.0237	6	94
1.6000	0.9300	0.0700	0.0255	7	93

Tied events

T=1.40 (tied events):

$$\begin{aligned}
 & (r_0 - q_0)/r_0 \cdot (r_{0.2} - q_{0.2})/r_{0.2} \cdot (r_{0.3} - q_{0.3})/r_{0.3} \cdot (r_{0.5} - q_{0.5})/r_{0.5} \cdot (r_{1.1} - q_{1.1})/r_{1.1} \cdot (r_{1.4} - q_{1.4})/r_{1.4} \\
 &= \hat{S}(1.1) \cdot (96 - 2)/96 \\
 &= 0.94
 \end{aligned}$$

Demo: Kaplan-Meier Survival Curve

- Censored events are marked by an asterisk (*):

The LIFETEST Procedure

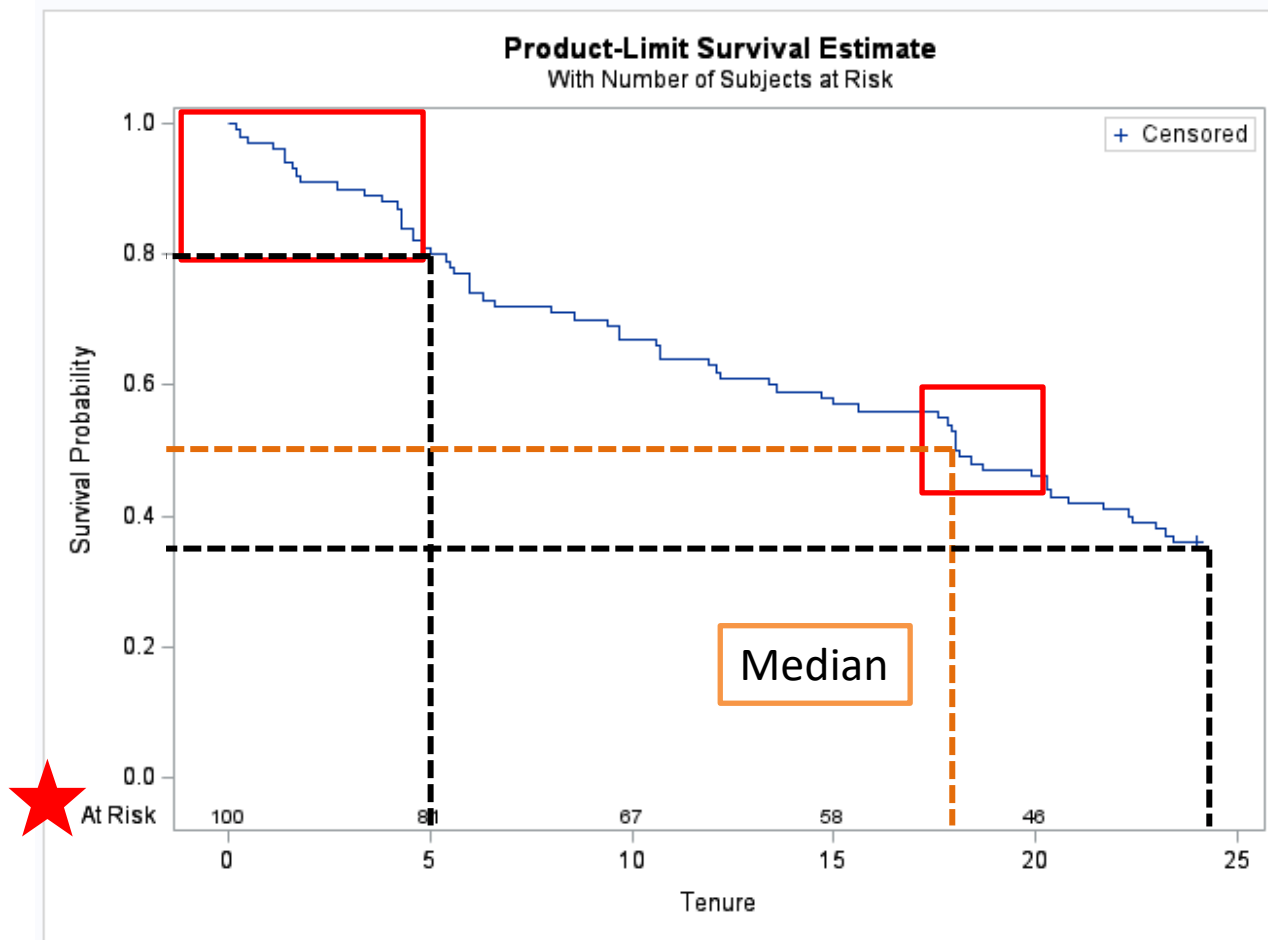
Product-Limit Survival Estimates						
Tenure		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	100
⋮						
23.2000		0.3700	0.6300	0.0483	63	37
23.4000		0.3600	0.6400	0.0480	64	36
24.0000	*	.	.	.	64	35
24.0000	*	.	.	.	64	34
24.0000	*	.	.	.	64	33

Censored observations

Note for each censored observation:

- Nr failed does not increase (not an event)
- Nr left (size of risk set) still decreases

Demo: Kaplan-Meier Survival Curve



Log-Rank Test

- Often want to **compare survival curves**:
 - Compare **survival rates by insurance marketing channel** to determine if clients lapse differently
 - Do customers that have signed up for the **loyalty programme** tend to **survive longer** than those that haven't?
 - Do **clients** with a **thinfile credit history** default sooner on their loans?
- Two-sample t -test or z -test can't be used due to **censoring**
- **Log-rank test** is a non-parametric test that is the most widely used method for comparing two or more survival curves

$$H_0: S_1(t) = S_2(t) = \dots = S_k(t)$$
$$H_1: S_i(t) \neq S_j(t) \text{ for at least one pair } (i, j)$$

Demo: Log-Rank Test

```
proc lifetest data=customer_churn
  conftype=asinsqrt
  plots=(survival (atrisk=0 to 24 by 5
test)) ; /* Add LR test p-val to KM */
  time tenure*status(0) ; /* Specify
censoring level */
  strata segment / test=logrank;

run;
```

- **Note:**

- `test` in `plots` adds log-rank test p-value to Kaplan-Meier plots
- `strata` argument added to specify groups to be compared
- `test=logrank` performs hypothesis test of survival equality

Demo: Log-Rank Test

- Kaplan-Meier estimated survival per group:

Stratum 1: Segment = Basic

Product-Limit Survival Estimates					
Tenure	Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000	1.0000	0	0	0	68
0.2000	0.9853	0.0147	0.0146	1	67
0.5000	0.9706	0.0294	0.0205	2	66

Stratum 2: Segment = Premium

Product-Limit Survival Estimates					
Tenure	Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000	1.0000	0	0	0	32
0.3000	0.9688	0.0313	0.0308	1	31
1.6000	0.9375	0.0625	0.0428	2	30

- Median survival time per group:

Summary Statistics for Time Variable Tenure

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper]
75	.	ASINSQRT	22.3000	.
50	17.7500	ASINSQRT	10.6000	20.8000
25	5.8000	ASINSQRT	4.3000	9.7000

Median survival time (Basic):

- 17.75 months
- Approx: 18 months

Summary Statistics for Time Variable Tenure

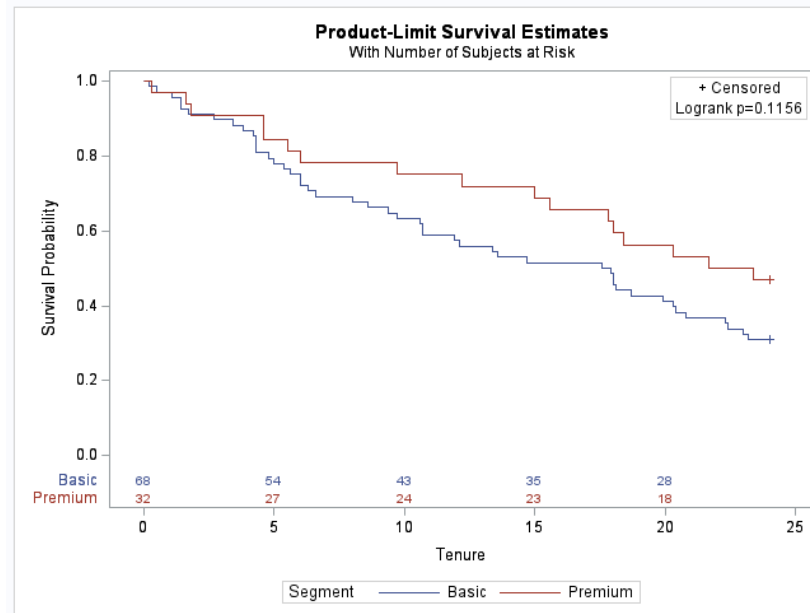
Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper]
75	.	ASINSQRT	.	.
50	22.5500	ASINSQRT	15.6000	.
25	10.9500	ASINSQRT	4.6000	18.4000

Median survival time (Premium):

- 22.55 months
- Approx: 23 months

Demo: Log-Rank Test

- Segment survival curves



- Hypothesis test output

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	2.4754	1	0.1156

- P-value=0.1156 > 0.05
- Do not reject H_0
- There is not sufficient statistically significant evidence to suggest that the survival curves differ by segment

What about Other Covariates?



Covariates could affect an observation unit's survival time or probability of survival



To investigate survival probability due to one factor, you must adjust for or **account for the effect of others**



This brings us to the application of **statistical modelling in survival analysis**

SURVIVAL MODELLING



Hazard Function

Cox Proportional Hazards Model

Proportional Hazards Assumption

Hazard Ratio

Hazard Function

- The **hazard** is the probability that an individual under observation at time t **experiences the event at time t**
- Represents **instantaneous event rate** for individual that survived to time t
 - **Example**: customer with company for 12 months, what is the probability that they leave in next instant (now)?

- **Define** hazard function as:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t | T \geq t)}{\Delta t}$$

- Provides vehicle for specifying **survival model**
 - closely **related** to the **survival function**
 - **modelling survival data** as a function of covariates relies on the hazard function

Hazard/Survival Function Relationship

- Recall **hazard function**:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t | T \geq t)}{\Delta t}$$

- Recall the formula of **conditional probability**:

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

$$\Rightarrow P(t \leq T \leq t + \Delta t | T \geq t) = \frac{P(t \leq T \leq t + \Delta t, T \geq t)}{P(T \geq t)}$$

The diagram shows the derivation of the hazard function formula using conditional probability. The numerator of the conditional probability formula is $P(t \leq T \leq t + \Delta t, T \geq t)$. This is annotated with a purple bracket labeled 'A' over $t \leq T \leq t + \Delta t$ and an orange bracket labeled 'B' over $T \geq t$. The denominator is $P(T \geq t)$, annotated with an orange bracket labeled 'B' over $T \geq t$. A blue star is placed to the left of the first equation.

- Applying this expression to $h(t)$:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t, T \geq t)}{P(T \geq t) \cdot \Delta t}$$

Hazard/Survival Function Relationship

- **Note:**

- $P(T \geq t)$: survival function $S(t)$
- For $t \leq T \leq t + \Delta t, T \geq t$
 - $P(t \leq T \leq t + \Delta t, T \geq t) = P(t \leq T \leq t + \Delta t)$

- The above remarks **simplify** $h(t)$ (slide 42) to:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t)}{\Delta t} \cdot \frac{1}{S(t)}$$

- From which **it follows** that:

$$h(t) = \frac{f(t)}{S(t)}$$

- This result shows the **relationship between the hazard function and the survival function**

Survival/Hazard Function Relationship

- Can be showed that: $h(t) = -\frac{d}{dt} \log S(t)$
- **Survival function** obtained by **integrating** the above expression:

$$-\int_0^t h(s)ds = \log S(t)$$

- Solving for the **survival function**:

$$S(t) = \exp\left(-\int_0^t h(s)ds\right) = e^{-H(t)}$$

- **Note**: $H(t)$ is called the **cumulative hazard**
- This result shows the **relationship between the survival function and the hazard function**
- **Why** is this relationship **so important**?
 - In CRM, we model hazard (risk factors) to predict survival (tenure)
 - If we want to change the survival curve, we need to adjust a risk factor, i.e., the hazard

Continuous-time Survival Analysis Approaches

- Kumar and Petersen apply continuous-time survival analysis
- Two approaches to continuous-time survival analysis:

1. Parametric models:

- Assume distribution of survival time is known
- Hazard function completely specified except for unknown parameters
- Use maximum likelihood to estimate parameters of interest (θ)
- E.g.: accelerated failure-time model (discuss in Retention later)

2. Semi-parametric models:

- Assume distribution of survival time is unknown
- Hazard function unspecified
 - No assumption about its functional form
 - We allow the instantaneous probability of event occurrence at time t , given survival until at least time t , to take any form
 - Hazard function is very flexible and can model a wide range of relationships between the covariates and survival time
- E.g.: Cox proportional hazards model

Cox Proportional Hazards Model

$$h(t|x_i) = \exp\left(\beta_0 + \sum_{j=1}^p \beta_j x_{ij}\right) = h_0(t) \cdot \exp\left(\sum_{j=1}^p \beta_j x_{ij}\right)$$

- $h_0(t) = \exp(\beta_0)$: (unspecified) **baseline hazard**
 - Hazard function for customer i when measured covariates $x_{i1}, \dots, x_{ip} = 0$
 - Function of time t , but not of the covariates
- $\exp\left(\sum_{j=1}^p \beta_j x_{ij}\right)$: **relative risk**
 - Risk of customer with covariates $x_i = [x_{i1} \ \dots \ x_{ip}]'$ experiencing event
 - **Relative (compared) to**: $x_i = [0 \ \dots \ 0]'$
 - Function of covariates but not t (covariates **do not change (much) over time**)

Demo: Cox Proportional Hazards Model

SAS code:

```
proc phreg data=customer_churn;  
  class Segment (ref='Basic') ;  
  model tenure*status(0) = Age Avg_Spend Segment ;  
run;
```

Model fit statistics & overall fit significance (***Use dataset with N=1000**):

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	7837.608	7749.754
AIC	7837.608	7755.754
SBC	7837.608	7768.949

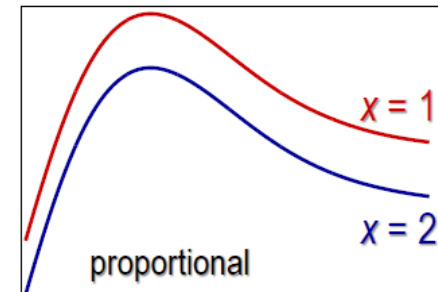
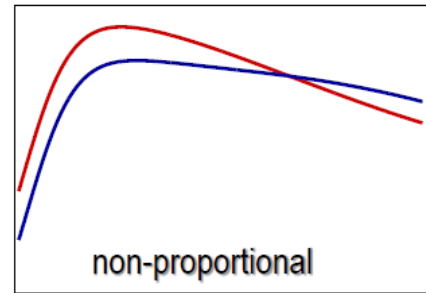
Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	87.8539	3	<.0001
Score	83.9996	3	<.0001
Wald	79.6712	3	<.0001

- Model with covariates is better
- AIC and SBC lower

- Overall model fit significant

Proportional Hazards Assumption

- It is called Cox proportional hazards model because of **proportional hazards (PH) assumption**
- PH assumption **requires**:
 - Hazard rate for one customer to be **proportional to** hazard rate of any other customer
 - Proportionality must be **constant over time**



- It is formally defined as:

$$\widehat{HR} = \frac{\hat{h}(y_i | \mathbf{x}_i^*)}{\hat{h}(y_i | \mathbf{x}_i)} = \hat{\theta}$$

- Hazard function for customer with covariates $\mathbf{x}_i^*$ is $\hat{\theta}$ times hazard function of customer with covariates $\mathbf{x}_i$
- Hazard function of one subject **proportional** to hazard function of another subject

Must be checked after fitting Cox PH model

Demo: Check Proportional Hazards Assumption

➤ Checking the proportional hazards (PH) assumption

H_0 : PH assumption **holds** (effect constant over time)

H_1 : PH assumption **violated** (effect changes over time)

```
proc phreg data=customer_churn;  
  class Segment (ref='Basic');  
  model tenure*status(0) = Age Avg_Spend Segment;  
  assess var=(Age Avg_Spend) ph / resample seed=12345;  
run;
```

Supremum Test for Proportionals Hazards Assumption				
Variable	Maximum Absolute Value	Replications	Seed	Pr > MaxAbsVal
Age	0.6098	1000	12345	0.7810
Avg_Spend	0.4378	1000	12345	0.9700
SegmentPremium	0.5974	1000	12345	0.7760

- All p-values > 0.05
- H_0 not rejected
- PH assumption holds

Proportional Hazards Assumption Violated

- Violation of the PH assumption does not mean Cox regression cannot be used
- It simply means that the model needs to be modified slightly to account for non-proportional hazards
- This is typically done by including an interaction between a predictor and time or by stratifying by a predictor

Cox PH Model Interpretation: Change Effect

- **Hazard ratio** ($\widehat{HR}$) used to interpret effect of change in significant covariate on hazard rate

$$\widehat{HR} = \frac{\hat{h}(y_i|\mathbf{x}_i^*)}{\hat{h}(y_i|\mathbf{x}_i)}$$

- $\hat{h}(y_i|\mathbf{x}_i^*)$: estimate of hazard rate for customer with measured covariates $\mathbf{x}^*$
- $\hat{h}(y_i|\mathbf{x}_i)$: estimate of hazard rate for customer with measured covariates $\mathbf{x}$
- This can be written as (see **slide 46** for model form)

$$\widehat{HR} = \frac{\hat{h}(y_i|\mathbf{x}_i^*)}{\hat{h}(y_i|\mathbf{x}_i)} = \frac{\cancel{\hat{h}_0(y_i)} \exp(\hat{\beta}_1 x_{i1}^* + \hat{\beta}_2 x_{i2}^* + \dots + \hat{\beta}_p x_{ip}^*)}{\cancel{\hat{h}_0(y_i)} \exp(\hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_p x_{ip})}$$

Cox PH Model Interpretation: Change Effect

- Estimated hazard ratio simplifies to

$$\widehat{HR} = \exp[\hat{\beta}_1(x_{i1}^* - x_{i1}) + \hat{\beta}_2(x_{i2}^* - x_{i2}) + \dots + \hat{\beta}_p(x_{ip}^* - x_{ip})]$$

- Estimated effect of change in x_j , holding all others constant, given by

$$\widehat{HR} = \exp[\hat{\beta}_j(x_{ij}^* - x_{ij})]$$

- If x_j is **numeric**:

- one unit change in $x_j \Rightarrow x_{ij}^* - x_{ij} = 1 \Rightarrow$ hazard ratio becomes

$$\widehat{HR} = \exp(\hat{\beta}_j)$$

- If x_j is **binary**:

- change in x_j between two subjects implies one is in class 1 while the other is in class zero $\Rightarrow x_{ij}^* - x_{ij} = 1 \Rightarrow$ hazard ratio also

$$\widehat{HR} = \exp(\hat{\beta}_j)$$

Demo: Interpreting the Change Effect

- From **proc phreg** output:

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
Age		1	-0.00469	0.00464	1.0186	0.3129	0.995	
Avg_Spend		1	-5.3294E-6	0.0003990	0.0002	0.9893	1.000	
Segment	Premium	1	-0.81371	0.09177	78.6287	<.0001	0.443	Segment Premium

- Only **Segment** is significant (p-value < 0.05)
 - Two levels: Basic (reference level) and Premium
 - Hazard rate for Premium customers is 0.443 times the hazard rate of Basic customers
 - Premium customers approximately 56% $[(1 - 0.44) * 100]$ less likely to churn at any given moment compared to Basic customers
- If **Age** were significant with $\widehat{HR} = 1.03$ (hypothetically...)
 - For every 1 year increase in Age, the risk of churn increases by 3% $[(1.03 - 1) * 100]$, holding all other variables constant

$$\begin{aligned}\widehat{HR} &= \exp(\hat{\beta}_j) \\ &= \exp(-0.81371) = 0.443\end{aligned}$$

Advantages/Disadvantages of the Cox PH Model

Advantages

- Does not require a particular probability distribution to represent survival times
- Relatively easy to incorporate time-dependent covariates
- Fairly robust to outliers
- Well studied and popular

Disadvantages

- Needs an estimate of the baseline hazard function for predictive modelling where you are not only interested in the relative effect of a covariate on the hazard rate
- Proportional hazards assumption not always valid
- Computationally expensive for big data (volume & number of covariates) where observations are made at discrete timepoints with many ties

Model Validation

- Validation is **not straightforward** because the outcome is **time-dependent** and data is often **censored**
- A **common approach** is to use **concentration curves** (similar to lift charts) to validate the hazard rate predictions
 1. Select a hypothetical scoring date and customers that have experienced an event prior to this date are excluded
 2. A validation interval is then chosen, and right censoring is applied
 3. Each customer is assigned a risk score ($X\hat{\beta}$) by the model and sorted from high to low based on this score
 4. A "hit" is registered if the event of interest is actually observed within the defined validation interval
 5. A concentration curve is created to visualise the percentage of "hits" captured within each decile of the scored data, showing how well the model concentrates high-risk customers at the top

Model validation demo: **AFT seminar**

Summary

Time-to-event data

Censoring

Survival function

Kaplan-Meier
Survival Curve

Log-Rank test

Hazard function

Cox Proportional
Hazards model

See Demo code for
NewCustomer
survival probability

See Demo code for
model validation

Homework



Classwork (in groups)

No classwork today!!
Review SAS demo file



Prep for 17 March

Kumar & Petersen Chapter 4
Sections 4.1 - 4.2 & 4.4 - 4.5